November 8, 2026 at 18:08

1. Introduction. The van der Waerden number $W(3,k)$ is the smallest n such that every string $x_1 \dots x_n$ of 0s and 1s has three equally spaced 0s or k equally spaced 1s; the clauses $waerden(3,k;n)$ are satisfiable exactly when $n < W(3,k)$. Exercise 7.2.2.2–6, on page 317, asks:

> **6.** [**HM37**] Use the Local Lemma to show that $W(3,k) = \Omega(k^2/(\log k)^3)$.

The answer on page 549 reads, in full:

> Let each x_i be 0 with probability $p = (2\ln k)/k$, and let n be at most $k^2/(\ln k)^3$. There are two kinds of "bad events": A_i, a set of three equally spaced 0s, occurs with probability $P = p^3$; and A'_j, a set of k equally spaced 1s, occurs with probability $P' = (1-p)^k \le \exp(-kp) = 1/k^2$. In the lopsidependency graph, which is bipartite, each A_i is adjacent to at most $D = 3k^3/((k-1)(\ln k)^3)$ nodes A'_j; each A'_j is adjacent to at most $d = \frac{3}{2}k^3/(\ln k)^3$ nodes A_i. By Theorem L, we want to show that, for all sufficiently large values of k, $P \le y(1-x)^D$ and $P' \le x(1-y)^d$, for some x and y.
>
> Choose x and y so that $(1-x)^D = 1/2$ and $y = 2P$. Then $x = \Theta((\log k)^3/k^2)$ and $y = \Theta((\log k)^3/k^3)$; hence $(1-y)^d = \exp(-yd + O(y^2 d)) = O(1)$. [See T. Brown, B. M. Landman, and A. Robertson, *J. Combinatorial Theory* **A115** (2008), 1304–1309.]

2. The answer is a sketch, and a sketch of this kind hides constants. This program brings them out.

1. *Events.* The probabilities P and P', and the claim that the lopsidependency graph is bipartite, are checked exactly on short strings: condition (133) of Lemma L by enumeration of all strings, and the rule of answer 351 for when two events must be adjacent.
2. *Degrees.* The largest numbers of neighbors are computed and compared with D and d.
3. *Numbers.* With the answer's choices, yd is exactly 24, so $(1-y)^d$ tends to e^{-24}. The second inequality then holds only when $(\ln k)^3 \ge 3e^{24}/\ln 2$, that is, when $\ln k$ exceeds about 4858. The program computes that threshold in logarithms, and also shows that the same argument with $p = (3\ln k)/k$ gives $W(3,k) > k^2/(81(\ln k)^2)$ for every $k \ge 8$.
4. *Concrete bounds.* For each k up to 30, the largest n for which Theorem L can be applied with the true degrees, compared with the known values of $W(3,k)$; and, for k up to 10^8, the largest n allowed by the answer's own D and d with free constants, compared with $k^2/(\ln k)^3$.
5. *Resampling.* Algorithm M finds strings with no bad event, and Theorem M bounds how often it resamples; both are checked.
6. *The solver.* The `sat` package recomputes $W(3,k)$ for $k \le 10$, and a plain backtracking search confirms the smallest values.

I wrote this on 16 September 2026.

3. Every claim is a mode.

```
package main
import (
    "context"
    "flag"
    "fmt"
    "math"
    "math/rand/v2"
    "os"
    "github.com/sjnam/sat"
)
⟨ Types 9 ⟩
⟨ Global variables 5 ⟩
⟨ Functions 6 ⟩
func main() {
    ⟨ Read the command line 4 ⟩
    if on("events") {
        ⟨ Check the events and the lopsidependency graph 12 ⟩
    }
    if on("degrees") {
        ⟨ Check the degree bounds D and d 21 ⟩
    }
    if on("numbers") {
        ⟨ Check the inequalities of Theorem L numerically 24 ⟩
    }
    if on("concrete") {
        ⟨ Compute the bounds that Theorem L gives for concrete k 28 ⟩
    }
    if on("resample") {
        ⟨ Run Algorithm M 30 ⟩
    }
    if on("solver") {
        ⟨ Recompute small values of W(3,k) 35 ⟩
    }
    ⟨ Say how it went 7 ⟩
}
```

4. ⟨ Read the command line 4 ⟩ ≡

```
flag.StringVar(&mode, "mode", "all",
    "events, degrees, numbers, concrete, resample, solver, or all")
flag.IntVar(&reps, "reps", 1000, "how many runs of Algorithm M")
flag.IntVar(&wmax, "wmax", 10, "the largest k for which W(3,k) is recomputed")
flag.Uint64Var(&seed, "seed", 20260916, "seed for the random choices")
flag.Parse()
```

This code is used in section 3.

5. ⟨ Global variables 5 ⟩ ≡

```
var (
    mode string
    reps, wmax int
    seed uint64
    failures int
)
```

This code is also defined in section 36.

This code is used in section 3.

6. Each claim is announced, checked, and marked; one `FAIL` anywhere makes the program exit with a nonzero status.

⟨ Functions 6 ⟩ ≡

```
func on(m string) bool { return mode ≡ m ∨ mode ≡ "all" }
func claim(ok bool, format string, args ... any) {
    mark := "ok␣␣"
    if ¬ok {
        mark = "FAIL"
        failures++
    }
    fmt.Printf("␣␣%s␣␣%s\n", mark, fmt.Sprintf(format, args...))
}
```

This code is also defined in sections 8, 10, 11, 16, 18, 19, 20, 22, 27, 32, 34, 39.

This code is used in section 3.

7. ⟨ Say how it went 7 ⟩ ≡

```
if failures > 0 {
    fmt.Printf("\n%d␣claims␣failed.\n", failures)
    os.Exit(1)
}
fmt.Println("\nEvery␣claim␣checked␣out.")
```

This code is used in section 3.

8. Progressions and events. Positions are $1, \ldots, n$. *aps* lists the arithmetic progressions of length L inside them, by increasing difference.

⟨ Functions 6 ⟩ +≡

```
func aps(n, L int) [][]int {
    var out [][]int
    for d := 1; (L − 1) ∗ d ≤ n − 1; d++ {
        for a := 1; a + (L − 1) ∗ d ≤ n; a++ {
            ap := make([]int, L)
            for t := range ap {
                ap[t] = a + t ∗ d
            }
            out = append(out, ap)
        }
    }
    return out
}
```

9. An event is a progression and a bit: the three-term progressions carry 0, for the events A_i, and the k-term ones carry 1, for the events A'_j.

⟨ Types 9 ⟩ ≡

```
type event struct {
    pos []int
    one bool
}
```

This code is used in section 3.

10. ⟨ Functions 6 ⟩ +≡

```
func events(k, n int) []event {
    var evs []event
    for _, ap := range aps(n, 3) {
        evs = append(evs, event{ap, false})
    }
    for _, ap := range aps(n, k) {
        evs = append(evs, event{ap, true})
    }
    return evs
}
```

11. A string of at most 62 bits is a bit vector, bit i for x_i.

⟨ Functions 6 ⟩ +≡

```
func happens(e event, x uint64) bool {
    for _, i := range e.pos {
        if (x ≫ i & 1 ≡ 1) ≠ e.one {
            return false
        }
    }
    return true
}
```

12. Events. With $x_i = 0$ with probability p, independently, an event A_i has probability p^3 and an event A'_j has probability $(1-p)^k$; and $(1-p)^k \le e^{-kp}$ because $1-p \le e^{-p}$, with $e^{-kp} = 1/k^2$ when $p = (2\ln k)/k$.

The claim that needs care is the graph. Lemma L on page 266 asks, in (133), that $\Pr(A_i \mid \bar{A}_{j_1} \cap \cdots \cap \bar{A}_{j_t}) \le p_i$ whenever none of $j_1, \ldots, j_t$ is a neighbor of i. The answer's graph joins A_i only to the A'_j that share a position with it, so events of the same kind are never neighbors even when they overlap. That is where "lopsided" comes in: three 0s are only made less likely by knowing that other triples are not all 0.

The program checks (133) by enumerating all 2^n strings with their exact probabilities, for $n \le 10$, $k = 3$ and 4, and three values of p. For every event it conditions on all its non-neighbors at once, and on 30 random subsets of them. There are at most 40 events, so the events that happen in a string fit in one word, and a set of events is a word too.

```
⟨ Check the events and the lopsidependency graph 12 ⟩ ≡
    fmt.Println("6. the events")
    ⟨ Check the probabilities of the events 13 ⟩
    r := rand.New(rand.NewPCG(seed, 6))
    var nCond, badCond int
    for _, k := range []int { 3, 4 }{
        for n := k; n ≤ 10; n++ {
            for _, p := range []float64 { 0.2, 0.5, 0.8 }{
                ⟨ Check condition (133) for all events on n positions 14 ⟩
            }
        }
    }
    claim(badCond ≡ 0, "%d conditional probabilities on up to 10 positions: " +
        "none exceeds Pr(A)", nCond)
    ⟨ Check the rule of answer 351 17 ⟩
```

This code is used in section 3.

13.

```
⟨ Check the probabilities of the events 13 ⟩ ≡
    badP := 0
    for k := 2; k ≤ 1000000; k = k * 3/2 + 1 {
        p := 2 * math.Log(float64(k))/float64(k)
        Pp := math.Pow(1 - p, float64(k))
        if p ≥ 1 ∨ Pp > math.Exp(-float64(k) * p) * (1 + 1e-12) ∨
            math.Abs(math.Exp(-float64(k) * p) * float64(k * k) - 1) > 1e-9 {
            badP++
        }
    }
    claim(badP ≡ 0, "(1-p)^k <= exp(-kp) = 1/k^2 when p = 2 ln k/k, for k " +
        "from 2 to a million")
```

This code is used in section 12.

14. The weight of a string is $\prod p^{[x_i=0]}(1-p)^{[x_i=1]}$. Two events are neighbors when one is an A and the other an A' and they share a position.

⟨ Check condition (133) for all events on n positions 14 ⟩ ≡

```
evs := events(k, n)
w, hap := make([]float64, 1 ≪ n), make([]uint64, 1 ≪ n)
for x := uint64(0); x < 1 ≪ n; x++ {
    w[x] = 1
    for i := 1; i ≤ n; i++ {
        if x ≪ 1 ≫ i & 1 ≡ 1 {
            w[x] *= 1 − p
        } else {
            w[x] *= p
        }
    }
    for b, e := range evs {
        if happens(e, x ≪ 1) {
            hap[x] |= 1 ≪ b
        }
    }
}
for a, ea := range evs {
    var others uint64
    for b, eb := range evs {
        if b ≠ a ∧ (ea.one ≡ eb.one ∨ ¬share(ea, eb)) {
            others |= 1 ≪ b
        }
    }
    ⟨ Compare the conditional probabilities of ea with its own 15 ⟩
}
```

This code is used in section 12.

15. ⟨ Compare the conditional probabilities of *ea* with its own 15 ⟩ ≡

```
alone := math.Pow(p, 3)
if ea.one {
    alone = math.Pow(1 - p, float64(k))
}
for trial := 0; trial ≤ 30; trial++ {
    sub := others
    if trial > 0 {
        sub &= r.Uint64()
    }
    var both, cond float64
    for x := uint64(0); x < 1 ≪ n; x++ {
        if hap[x] & sub ≡ 0 {
            cond += w[x]
            if hap[x] ≫ a & 1 ≡ 1 {
                both += w[x]
            }
        }
    }
    nCond++
    if cond > 0 ∧ both/cond > alone * (1 + 1e - 9) {
        badCond++
    }
}
```

This code is used in section 14.

16. ⟨ Functions 6 ⟩ +≡

```
func share(e, f event) bool {
    for _, i := range e.pos {
        for _, j := range f.pos {
            if i ≡ j {
                return true
            }
        }
    }
    return false
}
```

17. Algorithm M needs more than (133). Answer 351, due to Moser and Tardos, says when events i and j must be joined: when some setting makes A_i false and A_j true while some change to the variables of A_j would make A_i true, or the same with i and j exchanged. The program applies that rule literally, over all settings and all changes, for every ordered pair, and compares it with the answer's bipartite graph; since that graph is symmetric, one direction of the rule already has to match it.

⟨ Check the rule of answer 351 17 ⟩ ≡

```
var pairs, badRule int
for _, k := range []int { 3, 4 }{
    for n := k; n ≤ 9; n++ {
        evs := events(k, n)
        for a := range evs {
            for b := range evs {
                if a ≡ b {
                    continue
                }
                pairs++
                want := evs[a].one ≠ evs[b].one ∧ share(evs[a], evs[b])
                if mustJoin(evs[a], evs[b], n) ≠ want {
                    badRule++
                }
            }
        }
    }
}
claim(badRule ≡ 0, "%d ordered pairs of events on up to 9 positions: the rule " +
    "of answer 351 gives exactly the bipartite graph", pairs)
```

This code is used in section 12.

18. Is there a setting with A_i false and A_j true, and a change of the variables of A_j that makes A_i true?

⟨ Functions 6 ⟩ +≡

```
func mustJoin(ei, ej event, n int) bool {
    for x := uint64(0); x < 1 ≪ n; x++ {
        if happens(ei, x ≪ 1) ∨ ¬happens(ej, x ≪ 1) {
            continue
        }
        for c := 1; c < 1 ≪ len(ej.pos); c++ {
            y := x ≪ 1
            for t, i := range ej.pos {
                if c ≫ t & 1 ≡ 1 {
                    y ⊕= 1 ≪ i
                }
            }
            if happens(ei, y) {
                return true
            }
        }
    }
    return false
}
```

19. Degrees. An event A_i has three positions, and each position lies in at most $k\lfloor (n-1)/(k-1) \rfloor$ progressions of length k: it can be term $1, \ldots, k$, and the difference is at most $(n-1)/(k-1)$. So A_i has at most $3k(n-1)/(k-1)$ neighbors, which is below the answer's D when $n \le k^2/(\ln k)^3$. An event A'_j has k positions, and a position x lies in at most $\lfloor (n-x)/2 \rfloor + \min(x-1, n-x) + \lfloor (x-1)/2 \rfloor \le n-1$ progressions of length 3. So A'_j has at most $k(n-1)$ neighbors, below the answer's $d = 3kn/2$.

degrees counts the distinct neighbors exactly and returns the largest numbers, for the A's and for the A''s.

⟨ Functions 6 ⟩ +≡

```
func degrees(k, n int)(int, int) {
    three, long := aps(n, 3), aps(n, k)
    in3, inK := make([][]int, n + 1), make([][]int, n + 1)
    for i, ap := range three {
        for _, x := range ap {
            in3[x] = append(in3[x], i)
        }
    }
    for j, ap := range long {
        for _, x := range ap {
            inK[x] = append(inK[x], j)
        }
    }
    return most(three, inK, len(long)), most(long, in3, len(three))
}
```

20. The most distinct members of *through* met by one progression of *from*.

⟨ Functions 6 ⟩ +≡

```
func most(from [][]int, through [][]int, size int) int {
    best := 0
    seen := make([]int, size)
    for i, ap := range from {
        c := 0
        for _, x := range ap {
            for _, j := range through[x] {
                if seen[j] ≠ i + 1 {
                    seen[j], c = i + 1, c + 1
                }
            }
        }
        best = max(best, c)
    }
    return best
}
```

21. ⟨ Check the degree bounds D and d 21 ⟩ ≡

```
fmt.Println("6. the degrees")
var nd, badD, badd int
worstD, worstd := 0.0, 0.0
for k := 3; k ≤ 12; k++ {
    for n := k; n ≤ 100; n++ {
        D, d := degrees(k, n)
        nd++
        bD := 3 * float64(k) * float64(n − 1)/float64(k − 1)
        bd := float64(k) * float64(n − 1)
        if float64(D) > bD + 1e − 9 ∨ float64(D) > 3 * float64(k * n)/float64(k − 1) {
            badD++
        }
        if float64(d) > bd ∨ float64(d) > 1.5 * float64(k * n) {
            badd++
        }
        worstD, worstd = max(worstD, float64(D)/bD), max(worstd, float64(d)/bd)
    }
}
claim(badD ≡ 0, "%d pairs (k,n): A has at most 3k(n-1)/(k-1) <= D neighbors; " +
    "the largest ratio is %.3f", nd, worstD)
claim(badd ≡ 0, "A' has at most k(n-1) <= d neighbors; the largest ratio to " +
    "k(n-1) is %.3f", worstd)
```

This code is used in section 3.

22. Numbers. Write $L = \ln k$. The answer's choices make $y = 2p^3 = 16L^3/k^3$ and $d = \frac{3}{2}k^3/L^3$, so $yd = 24$ exactly, and $(1-y)^d = e^{-24}(1+O(y))$. That is a constant, which is what the argument needs; but the answer writes "$= O(1)$", an upper bound, when the step needs a lower bound. And the constant is small. Since $x \approx \ln 2/D \approx (\ln 2)L^3/(3k^2)$ and $P' \approx 1/k^2$, the inequality $P' \le x(1-y)^d$ becomes

$$1 \le \frac{\ln 2}{3} L^3 e^{-24}(1+o(1)), \qquad \text{or} \qquad L \ge \exp\bigl((24 + \ln 3 - \ln\ln 2)/3\bigr) \approx 4858.$$

So "sufficiently large" means $k > e^{4858}$, a number of 2110 digits.

margin evaluates $\ln\bigl(x(1-y)^d\bigr) - \ln P'$ for $p = cL/k$ and $n = ak^2/L^3$, with D and d exactly as in the answer and x and y chosen as there. Everything is done in logarithms, because k itself does not fit in a floating-point number.

```
⟨ Functions 6 ⟩ +≡
    func margin(L, c, a float64) float64 {
        lnp := math.Log(c) + math.Log(L) - L
        p := math.Exp(lnp)
        lnn := math.Log(a) + 2 * L - 3 * math.Log(L)
        ⟨ Compute ln P′, ln x, and ln(1 − y)^d 23 ⟩
        return lnx + lnpow - lnPp
    }
```

23. For large k the terms p, y, and $1/k$ underflow, so the series $\ln(1-t) = -t(1 + t/2 + \cdots)$ takes over where they are negligible.

```
⟨ Compute ln P′, ln x, and ln(1 − y)^d 23 ⟩ ≡
    lnPp := -c * L * (1 + p/2)
    if L < 30 {
        lnPp = math.Exp(L) * math.Log1p(-p)
    }
    lnD := math.Log(3) + lnn - math.Log1p(-math.Exp(-L))
    lnd := math.Log(1.5) + L + lnn
    lnt := math.Log(math.Ln2) - lnD
    lnx := lnt
    if lnt > -700 {
        lnx = math.Log(-math.Expm1(-math.Exp(lnt)))
    }
    lny := math.Ln2 + 3 * lnp
    y := math.Exp(lny)
    lnpow := -math.Exp(lnd + lny) * (1 + y/2)
    if y > 1e-8 {
        lnpow = math.Exp(lnd) * math.Log1p(-y)
    }
```

This code is used in section 22.

24. The program checks $yd = 24$, finds the threshold by bisection, compares it with the closed form, and checks that the margin changes sign only there, on a logarithmic grid of L from 1 to 10^6.

⟨ Check the inequalities of Theorem L numerically 24 ⟩ ≡

```
fmt.Println("6. the inequalities of Theorem L")
badYD := 0
for _, L := range []float64 { 2, 10, 100, 4858, 1e5 }{
    lnyd := math.Ln2 + 3 * (math.Log(2 * L) - L) + math.Log(1.5) + 3 * L - 3 * math.Log(L)
    if math.Abs(math.Exp(lnyd) - 24) > 1e - 9 {
        badYD ++
    }
}
claim(badYD ≡ 0, "yd = 24 exactly, so (1-y)^d tends to exp(-24) = %.3g",
    math.Exp(−24))
lo, hi := 1.0, 1e6
for i := 0; i < 200; i ++ {
    if mid := (lo + hi)/2; margin(mid, 2, 1) ≥ 0 {
        hi = mid
    } else {
        lo = mid
    }
}
closed := math.Exp((24 + math.Log(3) − math.Log(math.Ln2))/3)
claim(math.Abs(hi − closed) < 1e − 6 * closed, "the second inequality holds from " +
    "ln k = %.4f on, as the closed form %.4f predicts: k > 10^%.1f", hi, closed,
    hi/math.Ln10)
⟨ Check that the margin changes sign once 25 ⟩
⟨ Check the variant with p = (3 ln k)/k 26 ⟩
```

This code is used in section 3.

25. ⟨ Check that the margin changes sign once 25 ⟩ ≡

```
signs := 0
for i := 0; i ≤ 6000; i ++ {
    L := math.Pow(10, 6 * float64(i)/6000)
    if (margin(L, 2, 1) ≥ 0) ≠ (L ≥ hi) {
        signs ++
    }
}
claim(signs ≡ 0, "on 6001 values of ln k from 1 to 10^6 the margin is " +
    "negative below the threshold and positive above it")
```

This code is used in section 24.

26. Nothing in the argument forces $p = (2\ln k)/k$. With $p = (3\ln k)/k$ we have $P' \le 1/k^3$, and with $n = ak^2/L^3$ the product yd becomes $81a$; the second inequality then reads, up to lower-order terms, $81a + \ln a \le L + 3\ln L + \ln(\ln 2/3)$. Taking $a = L/81$, that is, $n = k^2/(81L^2)$, it holds for every L. So the same argument proves $W(3,k) = \Omega\big((k/\log k)^2\big)$, the bound published by Y. Li and J. Shu in *Advances in Applied Mathematics* **44** (2010), 243–247. The program checks the exact margin, on the same grid, from $k = 8$; below that $p = (3\ln k)/k$ is so large that $y = 2p^3$ is not a probability, or nearly so, and the margin is negative at $k \approx 7.7$.

⟨ Check the variant with $p = (3\ln k)/k$ 26 ⟩ ≡

```
badVar := 0
for i := 0; i ≤ 6000; i++ {
    L := math.Pow(10, 6 * float64(i)/6000)
    if L < math.Log(8) {
        continue
    }
    if ¬(margin(L, 3, L/81) ≥ 0) {
        badVar++
    }
}
claim(badVar ≡ 0, "with␣p␣=␣3␣ln␣k/k␣and␣n␣=␣k^2/(81␣(ln␣k)^2)␣both␣" +
    "inequalities␣hold␣for␣all␣k␣>=␣8␣on␣the␣grid:␣%d␣exceptions", badVar)
```

This code is used in section 24.

27. Concrete bounds. The answer's argument, like any Local Lemma argument, produces a definite bound for each k once the constants are free. *feasible* looks for p and x that satisfy Theorem L, with y as small as the first inequality allows, given the numbers D and d of neighbors; it searches logarithmic grids, so every success is a proof and a failure is only a failure to find one.

⟨ Functions 6 ⟩ +≡

```
func feasible(k, D, d float64)(bool, float64, float64, float64) {
    for ip := 1; ip < 300; ip++ {
        p := math.Pow(10, -9 + 9 * float64(ip)/300)
        lnPp := k * math.Log1p(-p)
        for ix := 1; ix < 300; ix++ {
            x := math.Pow(10, -16 + 16 * float64(ix)/300)
            lny := 3 * math.Log(p) - D * math.Log1p(-x)
            if lny ≥ 0 {
                break
            }
            y := math.Exp(lny)
            if lnPp ≤ math.Log(x) + d * math.Log1p(-y) {
                return true, p, x, y
            }
        }
    }
    return false, 0, 0, 0
}
```

28. For $k \le 30$ the degrees are the true ones, and the bound is compared with the table on page 189: $W(3,k)$ is known for $k \le 19$, and at least the tabulated number for $20 \le k \le 30$.

⟨ Compute the bounds that Theorem L gives for concrete k 28 ⟩ ≡

```
fmt.Println("concrete bounds")
badKnown := 0
var lllN [31]int
for k := 3; k ≤ 30; k++ {
    for n := k; ; n++ {
        D, d := degrees(k, n)
        if ok, _, _, _ := feasible(float64(k), float64(D), float64(d)); ¬ok {
            break
        }
        lllN[k] = n
    }
    if lllN[k] ≥ known[k] {
        badKnown++
    }
}
fmt.Printf("         k= 10, 20, 30: Theorem L gives W(3,k) > %d, %d, %d; " +
    "the table has %d, %d, %d\n", lllN[10], lllN[20], lllN[30], known[10],
    known[20], known[30])
claim(badKnown ≡ 0, "for 3 <= k <= 30 every such bound is below the table")
⟨ Compare the answer's constants with free ones for large k 29 ⟩
```

This code is used in section 3.

29. For large k the degrees are the answer's $D = 3kn/(k-1)$ and $d = \frac{3}{2}kn$, and the largest n is found by bisection. Even with p, x, and y free, these n stay below $k^2/(\ln k)^3$ as far as $k = 10^8$, though the ratio grows. They do exceed $k^2/(81(\ln k)^2)$, the variant above, which itself passes $k^2/(\ln k)^3$ once $\ln k > 81$, that is, once k has 36 digits.

⟨ Compare the answer's constants with free ones for large k 29 ⟩ ≡

```
var ratios []float64
badBig := 0
for e := 2; e <= 8; e++ {
    k := math.Pow(10, float64(e))
    if ok, _, _, _ := feasible(k, 3*k*k/(k-1), 1.5*k*k); !ok {
        fmt.Printf("        k= 10^%d: not even n= k is reached\n", e)
        continue
    }
    lo, hi := k, k*k
    for i := 0; i < 60; i++ {
        mid := math.Sqrt(lo * hi)
        if ok, _, _, _ := feasible(k, 3*k*mid/(k-1), 1.5*k*mid); ok {
            lo = mid
        } else {
            hi = mid
        }
    }
    L := math.Log(k)
    fmt.Printf("        k= 10^%d: n= %.4g, which is %.2f k^2/(ln k)^3 and " +
        "%.3f k^2/(ln k)^2\n", e, lo, lo*L*L*L/(k*k), lo*L*L/(k*k))
    if lo < k*k/(81*L*L) || len(ratios) > 0 && lo*L*L*L/(k*k) <= ratios[len(ratios)-1] {
        badBig++
    }
    ratios = append(ratios, lo*L*L*L/(k*k))
}
claim(badBig == 0 && len(ratios) > 0 && ratios[len(ratios)-1] < 1,
    "with the answer's D and d, the best n grows relative to k^2/(ln k)^3 but " +
        "stays below it, and above k^2/(81 (ln k)^2)")
```

This code is used in section 28.

30. Resampling. Algorithm M on page 266 sets every x_i to 1 with probability $\xi = 1 - p$, and while some event happens it resamples the positions of one such event. Theorem M says that if the conditions of Theorem L hold with numbers θ_j, the event A_j is resampled at most $\theta_j/(1-\theta_j)$ times on average. Here $\theta = y$ for the A's and $\theta = x$ for the A''s, with p, x, y from *feasible* at the largest n found for $k = 30$. The event resampled is always the first one that happens.

```
⟨ Run Algorithm M 30 ⟩ ≡
    fmt.Println("Algorithm M and Theorem M")
    k := 30
    n := k
    for {
        D, d := degrees(k, n + 1)
        if ok, _, _, _ := feasible(float64(k), float64(D), float64(d)); ¬ok {
            break
        }
        n++
    }
    D, d := degrees(k, n)
    _, p, x, y := feasible(float64(k), float64(D), float64(d))
    evs := events(k, n)
    r := rand.New(rand.NewPCG(seed, 30))
    var countA, countB, badString int
    for run := 0; run < reps; run++ {
        ⟨ Run Algorithm M once, counting resamples 31 ⟩
    }
    nA, nB := 0, 0
    for _, e := range evs {
        if e.one {
            nB++
        } else {
            nA++
        }
    }
    boundA, boundB := float64(nA) * y/(1 - y), float64(nB) * x/(1 - x)
    claim(badString ≡ 0, "k = 30, n = %d, p = %.3g: %d runs, each ending with no " +
        "three equally spaced 0s and no 30 equally spaced 1s", n, p, reps)
    claim(float64(countA)/float64(reps) ≤ boundA ∧ float64(countB)/float64(reps) ≤ boundB,
        "average resamples: %.3f for the %d A's (Theorem M: at most %.3f), %.3f for " +
            "the %d A's (at most %.3f)", float64(countA)/float64(reps), nA, boundA,
        float64(countB)/float64(reps), nB, boundB)
```

This code is used in section 3.

31. ⟨ Run Algorithm M once, counting resamples 31 ⟩ ≡

```
bits := make([]bool, n + 1)
for i := 1; i ≤ n; i++ {
    bits[i] = r.Float64() ≥ p
}
for {
    j := −1
    for t, e := range evs {
        if allEqual(bits, e) {
            j = t
            break
        }
    }
    if j < 0 {
        break
    }
    if evs[j].one {
        countB++
    } else {
        countA++
    }
    for _, i := range evs[j].pos {
        bits[i] = r.Float64() ≥ p
    }
}
⟨ Check the final string against the definition 33 ⟩
```

This code is used in section 30.

32. ⟨ Functions 6 ⟩ +≡

```
func allEqual(bits []bool, e event) bool {
    for _, i := range e.pos {
        if bits[i] ≠ e.one {
            return false
        }
    }
    return true
}
```

33. The check does not use the events: it looks at every difference.

⟨ Check the final string against the definition 33 ⟩ ≡

```
for a := 1; a ≤ n; a++ {
    for s := 1; a + 2 * s ≤ n; s++ {
        if ¬bits[a] ∧ ¬bits[a + s] ∧ ¬bits[a + 2 * s] {
            badString++
        }
    }
    for s := 1; a + (k − 1) * s ≤ n; s++ {
        ones := true
        for t := 0; t < k; t++ {
            ones = ones ∧ bits[a + t * s]
        }
        if ones {
            badString++
        }
    }
}
```

This code is used in section 31.

34. The solver. The clauses $waerden(3,k;n)$ say, for each three-term progression, that not all its positions are 0, and for each k-term progression, that not all are 1; here the variable x_i is true for a 1.

⟨ Functions 6 ⟩ +≡

```
func waerden(k, n int) *sat.Solver {
    s := sat.New()
    for i := 1; i ≤ n; i++ {
        s.NewVar()
    }
    for _, ap := range aps(n, 3) {
        s.AddClause(sat.Pos(ap[0]), sat.Pos(ap[1]), sat.Pos(ap[2]))
    }
    for _, ap := range aps(n, k) {
        c := make([]sat.Lit, k)
        for t, i := range ap {
            c[t] = sat.Neg(i)
        }
        s.AddClause(c...)
    }
    return s
}
```

35. For $3 \le k \le wmax$ the package should find a string of length $W(3,k)-1$, which is checked against the definition, and should find none of length $W(3,k)$. For $k \le 6$ a backtracking search that extends strings one bit at a time gives the values independently.

⟨ Recompute small values of $W(3,k)$ 35 ⟩ ≡

```
fmt.Println("W(3,k) with the sat package")
for k := 3; k ≤ wmax ∧ k ≤ 19; k++ {
    W := known[k]
    s := waerden(k, W − 1)
    st, _ := s.Solve(context.Background())
    good := st ≡ sat.Sat
    if good {
        ⟨ Check the model of s against the definition 37 ⟩
    }
    st2, _ := waerden(k, W).Solve(context.Background())
    claim(good ∧ st2 ≡ sat.Unsat, "W(3,%d) = %d: a checked string of length " +
        "%d, and none of length %d", k, W, W − 1, W)
}
⟨ Find W(3,k) by backtracking for k ≤ 6 38 ⟩
```

This code is used in section 3.

36. The table on page 189: $W(3,k)$ for $3 \le k \le 19$, then lower bounds for $20 \le k \le 30$.

⟨ Global variables 5 ⟩ +≡

```
var known = []int{3: 9, 18, 22, 32, 46, 58, 77, 97, 114, 135, 160, 186, 218,
    238, 279, 312, 349, 389, 416, 464, 516, 593, 656, 727, 770, 827, 868, 903}
```

37. ⟨ Check the model of s against the definition 37 ⟩ ≡

```
bits := make([]bool, W)
for i := 1; i < W; i++ {
    bits[i] = s.Value(sat.Pos(i))
}
for _, ap := range aps(W − 1, 3) {
    good = good ∧ (bits[ap[0]] ∨ bits[ap[1]] ∨ bits[ap[2]])
}
for _, ap := range aps(W − 1, k) {
    ones := true
    for _, i := range ap {
        ones = ones ∧ bits[i]
    }
    good = good ∧ ¬ones
}
```

This code is used in section 35.

38. The search extends a string by one bit at a time and checks only the progressions that end at the new bit; the longest string it can reach is $W(3, k) − 1$.

⟨ Find $W(3, k)$ by backtracking for $k \le 6$ 38 ⟩ ≡

```
for k := 3; k ≤ 6; k++ {
    bits := make([]bool, 64)
    longest := 0
    var extend func(n int)
    extend = func(n int) {
        longest = max(longest, n)
        for _, b := range []bool { false, true }{
            bits[n + 1] = b
            if ¬endsBad(bits, n + 1, k) {
                extend(n + 1)
            }
        }
    }
    extend(0)
    claim(longest + 1 ≡ known[k], "backtracking: the longest good string for " +
        "k = %d has length %d", k, longest)
}
```

This code is used in section 35.

39. Does some progression ending at position m have three 0s or k 1s?

⟨ Functions 6 ⟩ +≡

```
func endsBad(bits []bool, m, k int) bool {
    for s := 1; m − 2 ∗ s ≥ 1; s++ {
        if ¬bits[m] ∧ ¬bits[m − s] ∧ ¬bits[m − 2 ∗ s] {
            return true
        }
    }
    for s := 1; m − (k − 1) ∗ s ≥ 1; s++ {
        ones := true
        for t := 0; t < k; t++ {
            ones = ones ∧ bits[m − t ∗ s]
        }
        if ones {
            return true
        }
    }
    return false
}
```

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